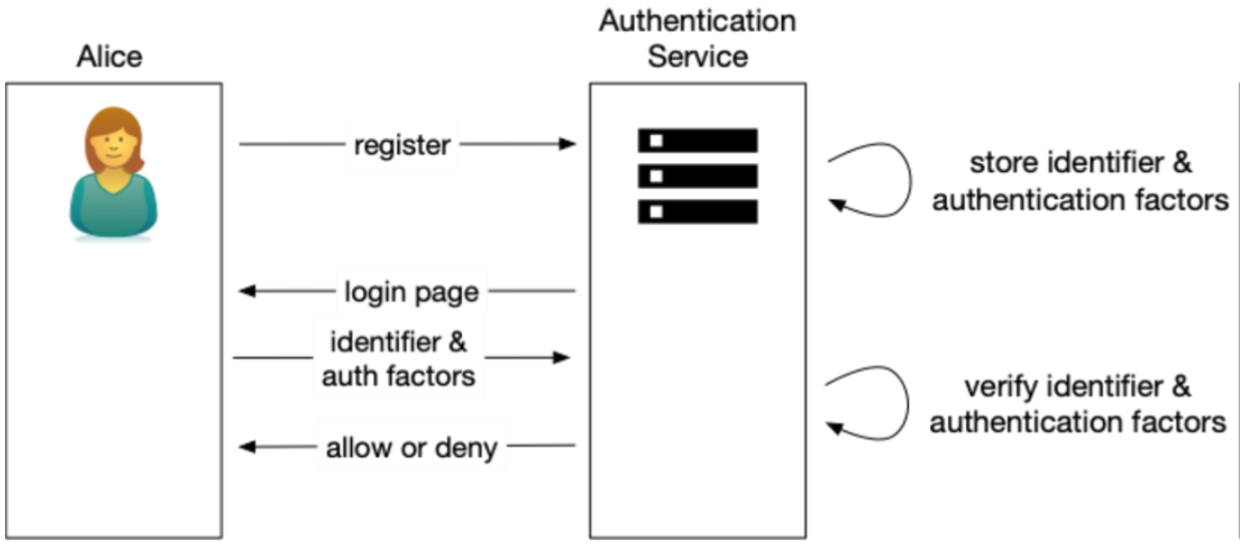


Figure 3.7 Identifier with authentication factors flow. During registration, Alice provides an identifier (such as a username or email) and one or more authentication factors (like a password or biometric). The authentication service securely stores these. At login, Alice is prompted for her identifier and corresponding authentication factors. The service verifies both before granting or denying access. This is the most common authentication pattern and serves as the foundation for multi-factor authentication, where the combination of factors enhances security by requiring control over multiple types of evidence.

When registering, Alice provides an identifier, chooses a password, and, usually, chooses what additional authentication factors she wants to use. The authentication service binds these authentication factors to the identifier, storing them securely. Alice must provide these each time she logs in.

3.6.4 Challenge-response

In challenge-response, after Alice provides an identifier to indicate which account she's claiming, the system sends back a challenge, and Alice must send back the correct response to prove she has control over a secret. Unlike a password, which is static, the challenge and response are usually derived in real time, allowing Alice to prove control over some secret without revealing it.

A *CAPTCHA* is a kind of challenge-response authentication method, although it's not usually used to prove that a specific requester knows a specific secret—rather, it's a challenge where the response is supposed to prove the requester is human. Still, it meets the description from the